# Refractometer



Category: **Laboratory Instruments & Glassware** Type: **Short SOP** Priority: **Medium**

4 labs

**Covers these lab items:** Hand Refractometer; Refractometer

## What it is used for

To measure the refractive index or concentration of a suitable liquid using a refractometer. To obtain accurate results by calibrating the instrument and preparing the sample correctly.

## Main parts

- Prism or sample surface
- Cover plate
- Eyepiece or digital display
- Calibration control
- Sample dropper

## Before you start

1. Check the refractometer type and measurement range for the sample.
2. Clean and dry the prism surface.
3. Calibrate using the specified reference solution or distilled water as instructed.
4. Mix the sample gently and bring it to the required temperature if specified.
5. Check that the scale or display is readable.

## How to use

1. Place a small amount of sample on the prism surface.
2. Close the cover plate gently if the model has one.
3. Aim the instrument toward a suitable light source if required for an optical model.
4. Focus the eyepiece and read the boundary line, or read the digital display.
5. Apply temperature correction if required by the method.
6. Record the result and sample details.
7. Clean the prism immediately after measurement using an approved method.
8. Store the instrument safely.

## ⚠ Safety

- Follow sample-specific precautions, especially for corrosive or hazardous liquids.
- Do not look directly at the sun through an optical instrument.
- Avoid scratching the prism surface.
- Use clean droppers to prevent cross-contamination.

## Care & storage

- Clean and dry the prism after each sample.
- Store with the cover closed and protective case fitted.
- Check calibration regularly.
- Avoid dropping or exposing the instrument to excessive heat.

## Common mistakes to avoid

- Skipping calibration.
- Using a dirty prism or contaminated dropper.
- Reading before focusing correctly.
- Ignoring required temperature correction.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*